Midterm Exam Review

- Descriptive Statistics and Data Classification
- Random Variables and Probability
- Sampling Distributions and Properties of Sample Mean
- ANOVA
- t Test
- z Test
- Chi-squared test
- Measures of Association for Nominal Variables
- R Code and Outputs

Midterm Exam: 90 minutes

Monday, 10:00 to 11:30 AM

You should bring the following items:

1) TWO 8.5" x 11" double-sided sheets of notes (anything you wish to put)
2) All statistical tables (WITHOUT notes)
   You can find them on the course's Brightspace page
   No tables will be distributed during the exam
   (z table – Normal Dist., F table, t table, Chi Sq. table)

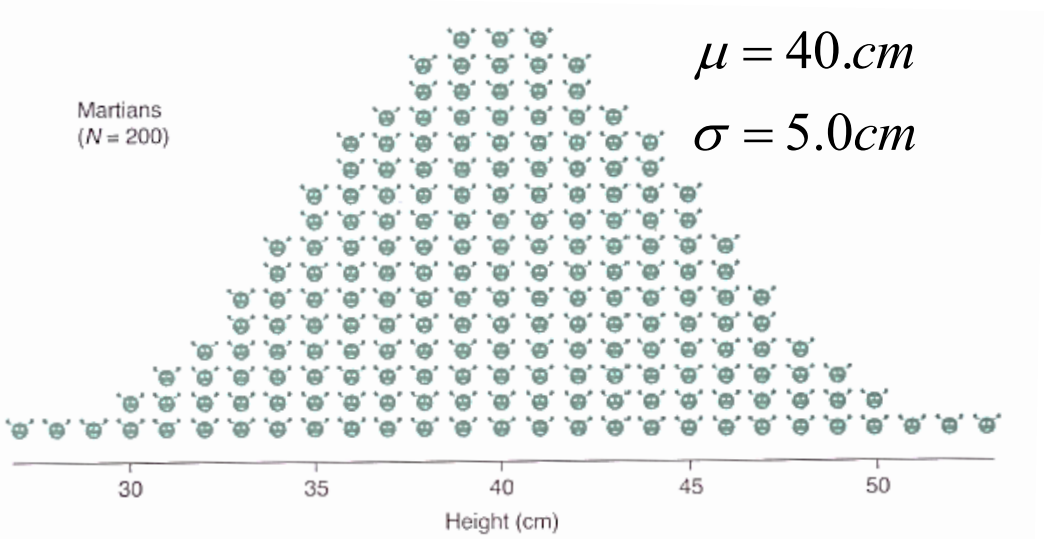
3) Calculator (a phone is not an acceptable calculator)

All material through Session 13 (Descriptive Statistics through Measures of Association for Nominal Variables); not Session 14 lecture on Types of Errors/Power of Test

Review
- Reading from textbook (see syllabus)
- Lecture slides
- HW assignments
- Other questions at end of chapters

Exam: 6-8 Questions, 2-3 on probability

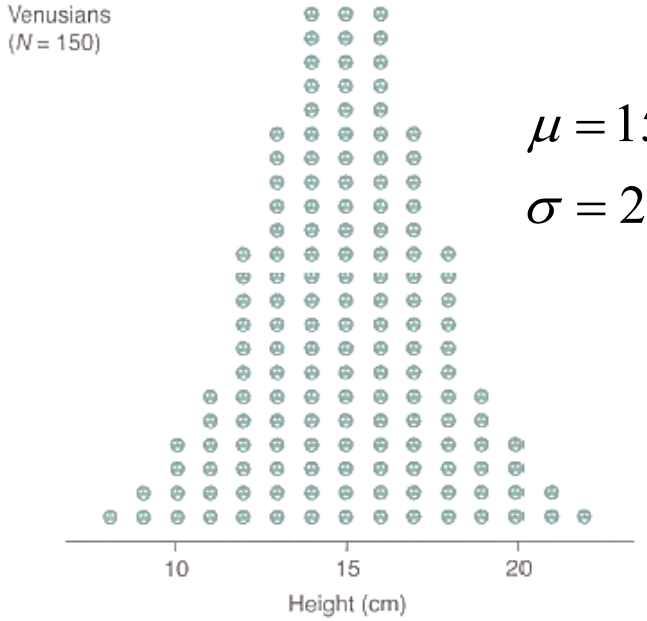
Population Statistics



$\mu = 40.cm$

$\sigma = 5.0cm$

FIGURE 2-1. Distribution of heights of 200 Martians, with each Martian's height represented by a single point. Notice that any individual Martian is more likely to have a height near the mean height of the population (40 cm) than far from it and is equally likely to be shorter or taller than average.



$\mu = 15cm$

$\sigma = 2.5cm$

FIGURE 2-2. Distribution of heights of 150 Venusians. Notice that although the average height and dispersion of heights about the mean differ from those of Martians (Fig. 2-1), they both have a similar bell-shaped appearance.

$$x_i \ , \ i = 1, \ldots, \ N$$

Central Tendency - Mean

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Dispersion – Variance $\left(\sigma^2\right)$ or Standard Deviation $\left(\sigma\right)$

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}\left(x_i - \mu\right)^2$$

Estimates from a Limited Sample – Sample Statistics

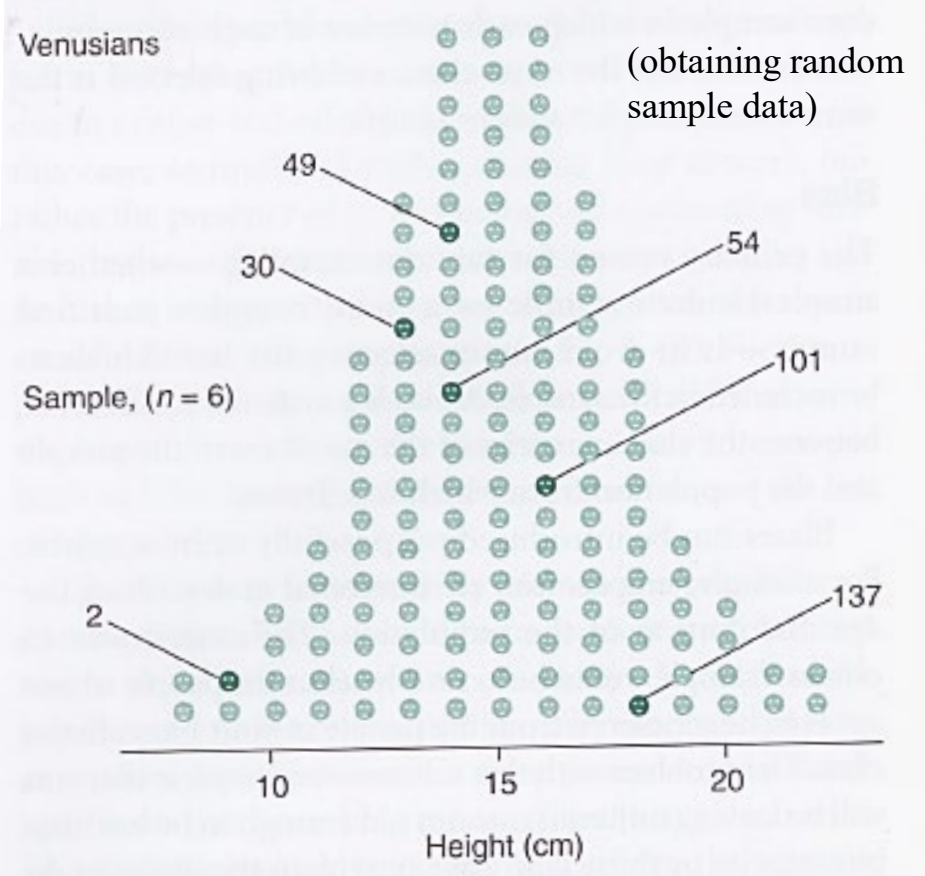


FIGURE 2-3. To select $n = 6$ Venusians at random, we number the entire population of $N = 150$ Venusians from 1 to 150, beginning with the first individual on the far left of the population as number 1. We then select six random numbers from Table 2-2 and select the corresponding individuals for the sample to be observed.

Goal of random sampling is to avoid bias because we want to use sample statistics as estimates for population statistics

$$x_i \ , \ i = 1, \ldots, \ n \qquad n < N$$

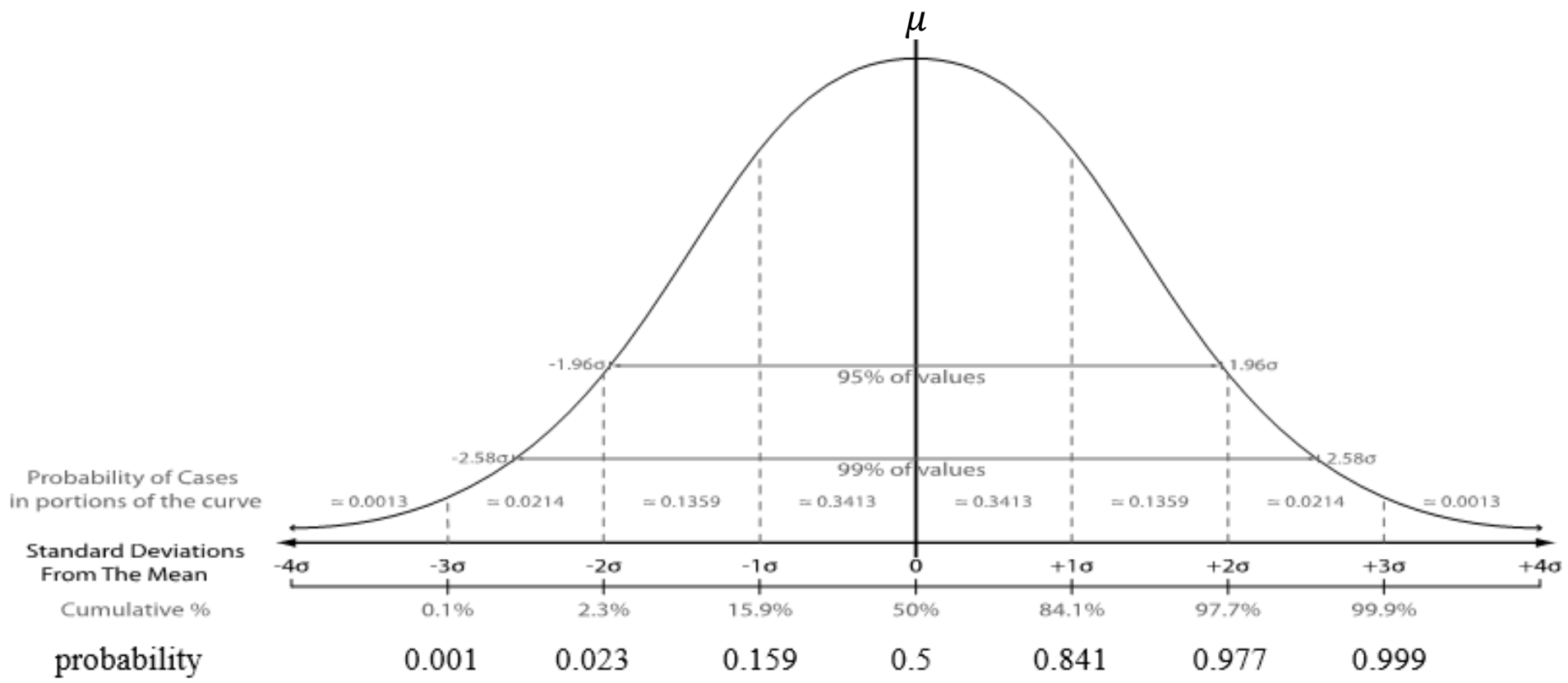
Central Tendency – Sample Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Dispersion – Sample Variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(x_i - \bar{x} \right)^2$$

Quantiles of the Normal Distribution



probability	0.001	0.023	0.159	0.5	0.841	0.977	0.999

Some Measures of Central Tendency and Dispersion

Sort the data in ascending order first:

$$x_i \ , \ i = 1, \ldots, \ n \qquad x_1 \le x_2 \ldots \le x_n$$

Central Tendency – Median

For *n* odd: Median = $x_j \ , \quad j = (n+1)/2$

For *n* even: Median = $\dfrac{x_j + x_k}{2} \ , \quad j = \dfrac{n}{2}, \quad k = \dfrac{n}{2} + 1$

Dispersion – Range (smallest and largest values) $\left[x_1, \quad x_n \right]$

Dispersion – Interquartile Range (IQR)

IQR = 75th percentile – 25th percentile of the data

Calculating Percentiles from Sample Data

p^{th} percentile of a data set is the smallest number x such that p% of the data is less than x

1. Sort the data in ascending order:

$$x_i, i = 1, \ldots, n \qquad x_1 \le x_2 \ldots \le x_n$$
$$1 \qquad 2 \ldots \qquad n$$

2. Calulate rank or index $== \dfrac{p(n-1)}{100} + 1 = I_R + F_R$

Integer part of answer = I_R = integer rank; Fraction part of answer = F_R

3. Calculate p^{th} percentile by interpolation:

Find observations with ranks I_R & $I_R + 1$ $\left(x_{I_R} \ \& \ x_{I_R+1}\right)$

$$p^{th} \text{ percentile} = x_{I_R} + F_R \cdot \left(x_{I_R+1} - x_{I_R}\right)$$

Classification of Data

Interval Data (measurement scale has constant intervals)

- Continuous (can take on any value within a range)
  - e.g., weight, pH, LDL, blood pressure, height
- Discrete (limited to a set of values)
  - e.g., # tender joints, # asthmatic episodes

Categorical Data (measurement scale is set of categories)

- Ordinal (natural ordering to the data)
  - e.g., cancer stage (I,II,III,IV),
    pain level (none, mild, moderate, strong, severe)
- Nominal (unordered scale)
  - e.g., sex (F, M), cause of death (accident, heart disease, etc.)

Discrete Random Variable

Definition: RV that can take on only one of a finite set of values

Example 1: Outcome of a coin toss – H or T

Represent the RV by X and its value by x

	H	T
x	1	2

Prob $(X = x)$?

Prob $(X = 1) = 1/2$

Prob $(X = 2) = 1/2$

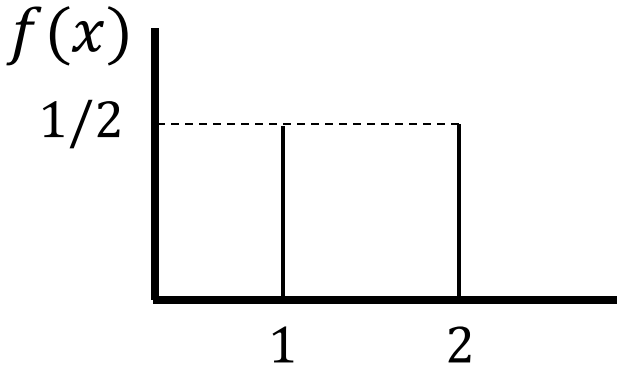
Probability Function (p.f.)

Let X denote a discrete RV with values: $x_1, x_2, \ldots, x_k$

The Probability Function of X is denoted $f(x)$, where

$$f(x) = \text{Prob}(X = x)$$

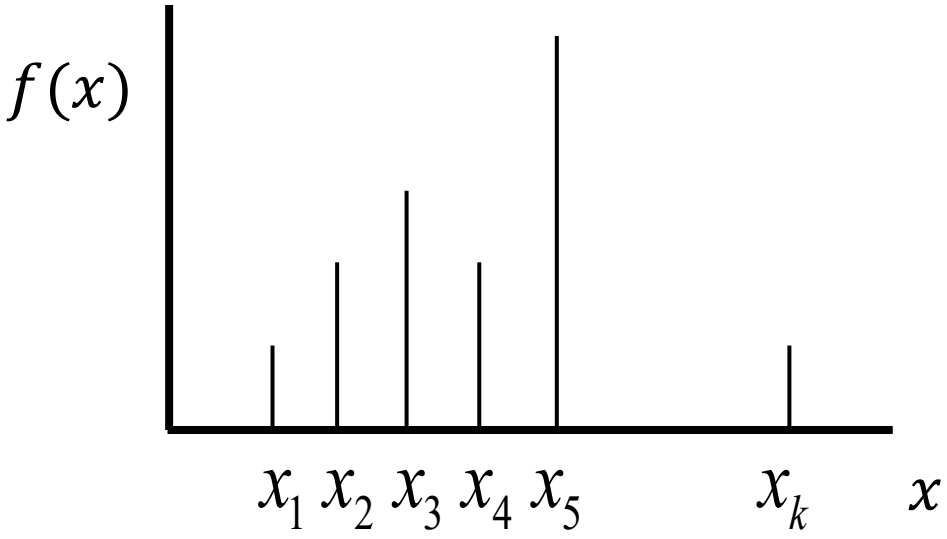
Example 1: Outcome of a coin toss – H $(X = x_1 = 1)$ or T $(X = x_2 = 2)$



$$f(x) = 1/2, \quad x = 1, \; x = 2$$
$$0, \quad \text{elsewhere}$$

Properties of Discrete Probability Function

$$f(x) = \text{Prob}(X = x)$$



Properties

$$f(x_i) \geq 0$$

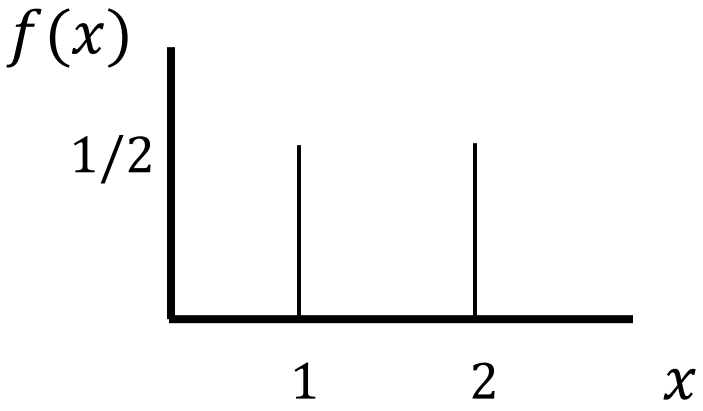
$$\sum_{i=1}^{k} f(x_i) = 1$$

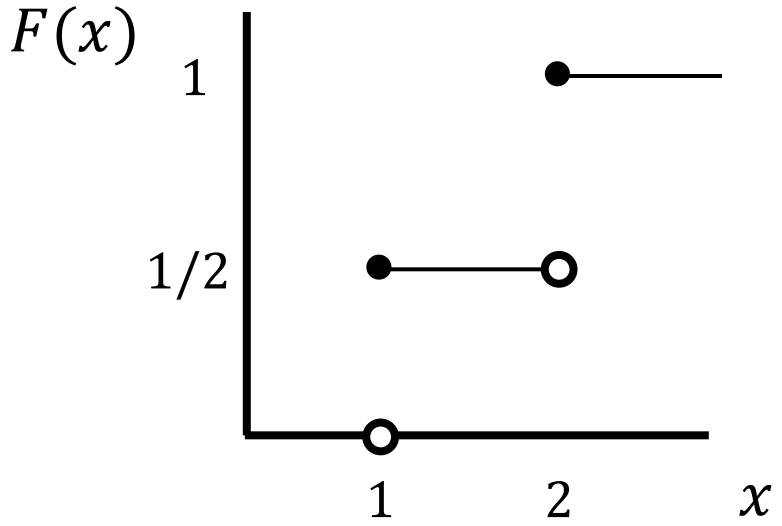
$f(x_i)$ $\quad x_1, x_2, ..., x_k$

$f(x) = 0$ elsewhere

Distribution Function (d.f.)

$$f(x) = \text{Prob}(X = x) \qquad F(x) = \text{Prob}(X \le x) = \sum_{y:y \le x} f(y)$$





Others: e.g., d.f. for loaded die RV $f(x) = 0.20, \quad x = 1$

$f(x) = 0.16, \quad x = 2, \ldots, 6$

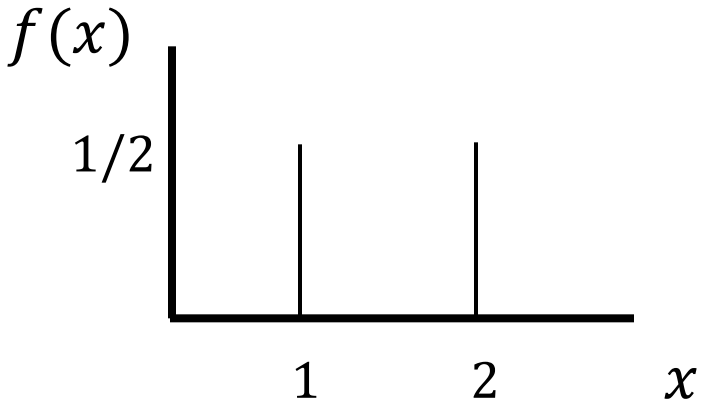
$0, \qquad$ elsewhere

Expected Value of X (mean of X)

Let X denote a discrete RV with values in set D and p.f. $f(x)$

$$E[X] = \mu = \sum_{x \in D} x \cdot f(x) = \frac{1}{N} \sum_{i=1}^{N} x_i$$

Example



$$E[X] = \sum_{x \in D} x f(x)$$

$$= 1\left(\tfrac{1}{2}\right) + 2\left(\tfrac{1}{2}\right) = 1.5$$

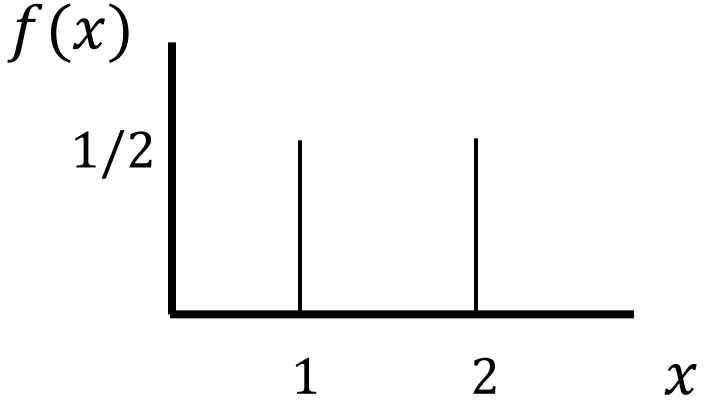
Others: e.g., loaded die

Expected Value of a Function of X

Let X denote a discrete RV with values in set D and p.f. $f(x)$

Define a function of X as $h(X)$

$$E[h(X)] = \sum_{x \in D} h(x) \cdot f(x) = \frac{1}{N} \sum_{i=1}^{N} h(x_i)$$



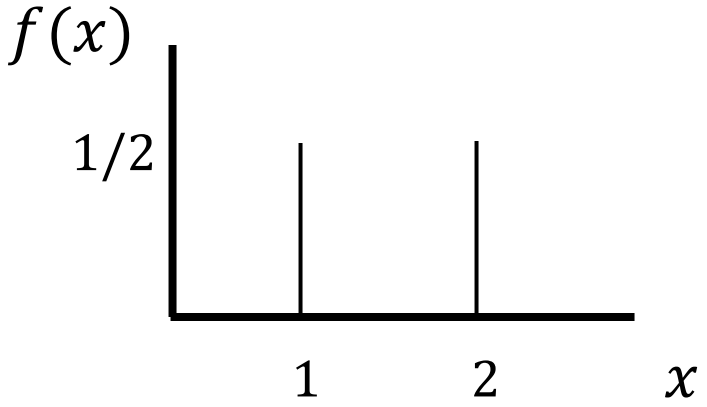
Examples

$$h(X) = X^2$$

$$h(X) = \left(X - E[X]\right)^2 = \left(X - 1.5\right)^2$$

For each, find: $E[h(X)]$

Examples



Find: $E[h(X)]$

$$h(X) = X^2$$

$$h(X) = (X - E[X])^2$$

$$h(X) = X^2: \quad E[X^2] = \sum_{x \in D} x^2 f(x) = 1^2\left(\tfrac{1}{2}\right) + 2^2\left(\tfrac{1}{2}\right) = 2.5$$

$$h(X) = (X - E[X])^2: \quad E\left[(X - E[X])^2\right] = E\left[(X - 1.5)^2\right]$$

Variance of X

$$= \sum_{x \in D} (x - 1.5)^2 f(x)$$

$$= (-0.5)^2\left(\tfrac{1}{2}\right) + (0.5)^2\left(\tfrac{1}{2}\right) = 0.25$$

Probability Density Function (Continuous RV)

How to describe the "degree of randomness" of a continuous RV X?

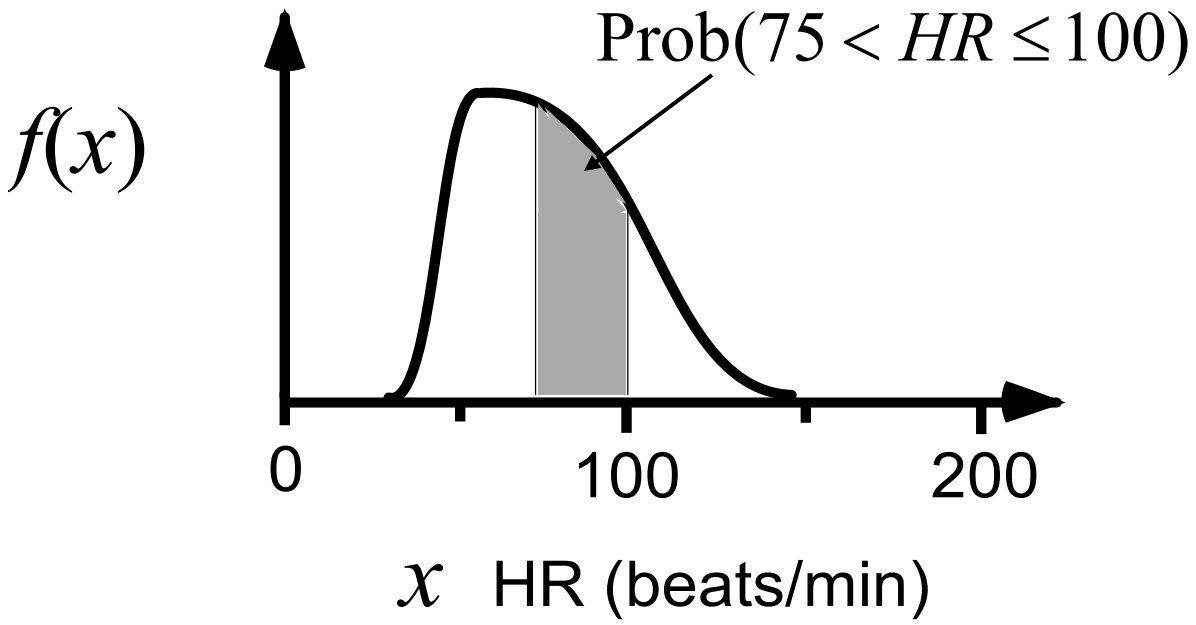
Probability density function (p.d.f.): $f(x)$

$$\text{Prob}(a < X \le b) = \int_a^b f(x)dx$$

Properties

$$f(x) \ge 0$$

$$\int_{-\infty}^{\infty} f(x)dx = 1$$



Expected Value

Let X denote a continuous RV with $X \sim f(x)$

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx = \mu$$

Define a function of X as $h(X)$

$$E[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) dx$$

Distribution Function (d.f.)

Let RV $X \sim f(x)$ probability density function

$F(x)$ distribution function

$$F(x) = \text{Prob}(X \le x) = \Pr(X \le x)$$

$$\boldsymbol{F(x) = \int_{-\infty}^{x} f(u) \cdot du}$$

Relationship between p.d.f. and d.f.

Properties

$0 \le F(x) \le 1 \quad -\infty < x < \infty$

$F(-\infty) = 0 \quad F(\infty) = 1$

$F(x)$ nondecreasing with increasing x

Other Relationships Involving d.f.

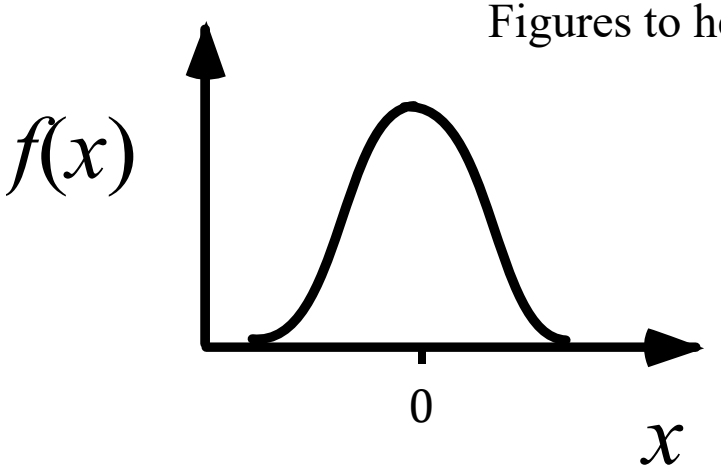
Recall: $F(x) = \Pr(X \le x)$

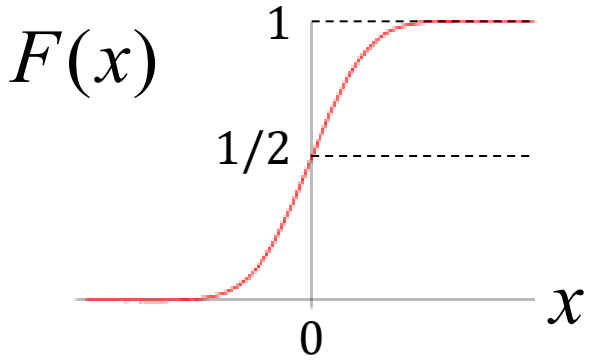
$$\boldsymbol{F(x) = \int_{-\infty}^{x} f(u) \cdot du}$$

$$\Pr(X > x) = 1 - F(x)$$

$$\Pr(x_1 < X \le x_2) = F(x_2) - F(x_1) = \int_{x_1}^{x_2} f(x)\,dx$$

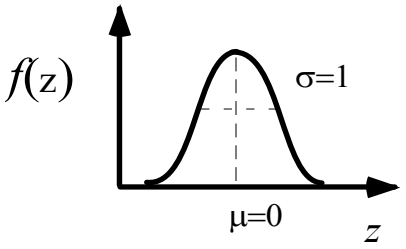
Figures to help illustrate these ideas





Standard Normal RV *Z*

$Z \sim N(0, 1^2)$



$$f(z) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-(z)^2}{2}\right)$$

p.d.f.

d.f. → $F(z) = \int_{-\infty}^{z} \frac{1}{(2\pi)^{1/2}} \exp\left(-\frac{1}{2}u^2\right) du$

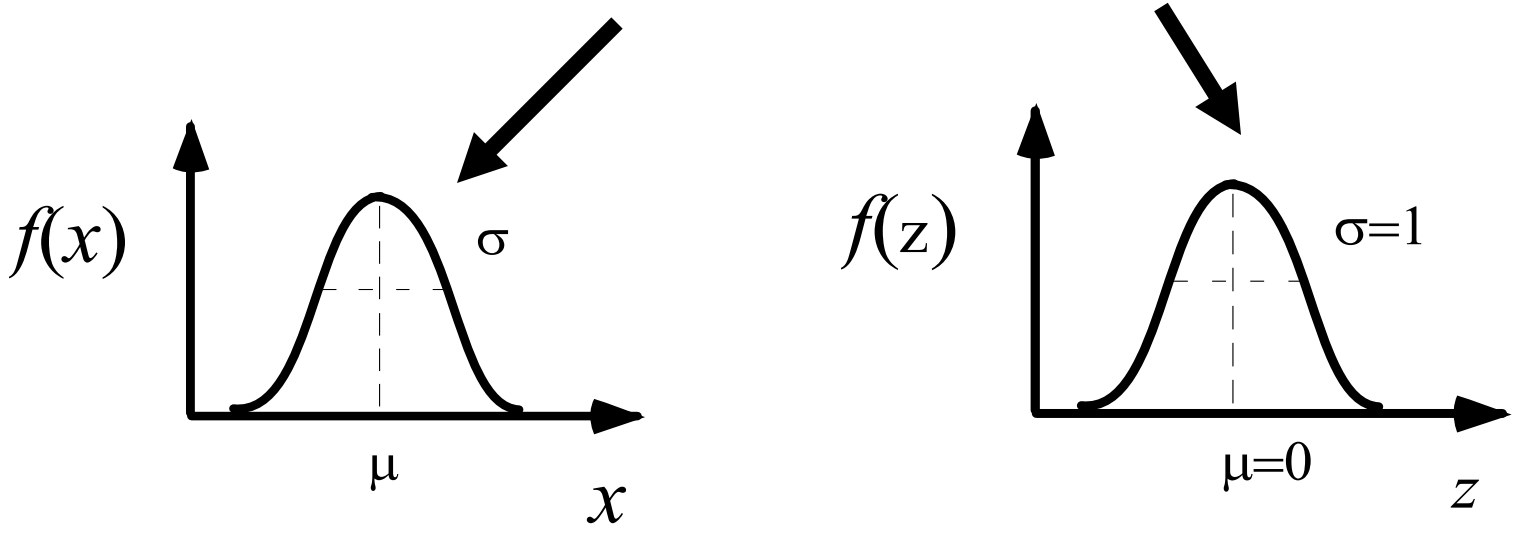
Table of the Standard Normal Distribution Function

z	$F(z)$
0.00	0.5000
0.01	0.5040
0.02	0.5080
0.03	0.5120
0.04	0.5160
0.05	0.5199

z	$F(z)$
3.55	0.9998
3.60	0.9998
3.65	0.9999
3.70	0.9999
3.75	0.9999
3.80	0.9999
3.85	0.9999
3.90	1.0000

Normal RV and Standard Normal RV

Relationship between $X \sim N(\mu, \sigma^2)$ and $Z \sim N(0, 1^2)$

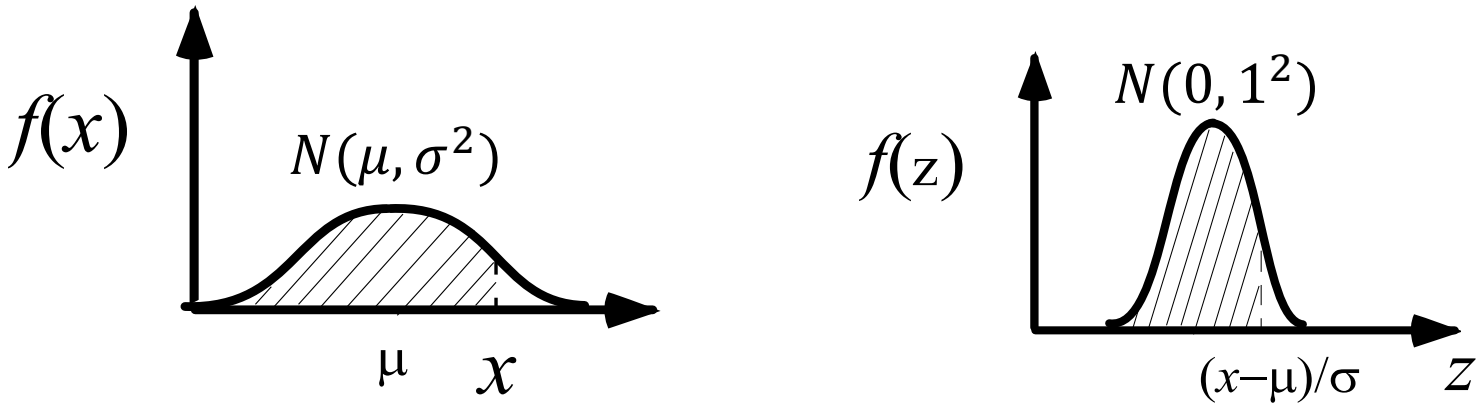


Define $x = \mu + \sigma z$

therefore $z = \dfrac{x - \mu}{\sigma}$

The following can be shown:

$$\Pr(X \le x) = \Pr\left(Z \le \frac{x-\mu}{\sigma}\right)$$



Therefore, the following are also true (for example):

$$\Pr(X > x) = 1 - \Pr(X \le x) = 1 - \Pr\left(Z \le \frac{x-\mu}{\sigma}\right)$$

$$\Pr(a < X \le b) = \Pr(X \le b) - \Pr(X \le a) = \Pr\left(Z \le \frac{b-\mu}{\sigma}\right) - \Pr\left(Z \le \frac{a-\mu}{\sigma}\right)$$

What is a Sampling Distribution?

- Any statistic calculated from sample data is another random variable, with its own characteristic distribution
- For example, the sample mean has a probability distribution: how the statistic varies in value across all sample groups of size n that might be selected
- If you choose 25 sample groups of size n, you will get 25 different sample means
- Each of these sample means is an estimate of the population mean and the distribution of all of the possible values of the sample mean is called a **sampling distribution**

Sample Statistics: How well do they estimate Population Statistics?

$\bar{x}$ is an estimate of μ $\quad$ s is an estimate of σ

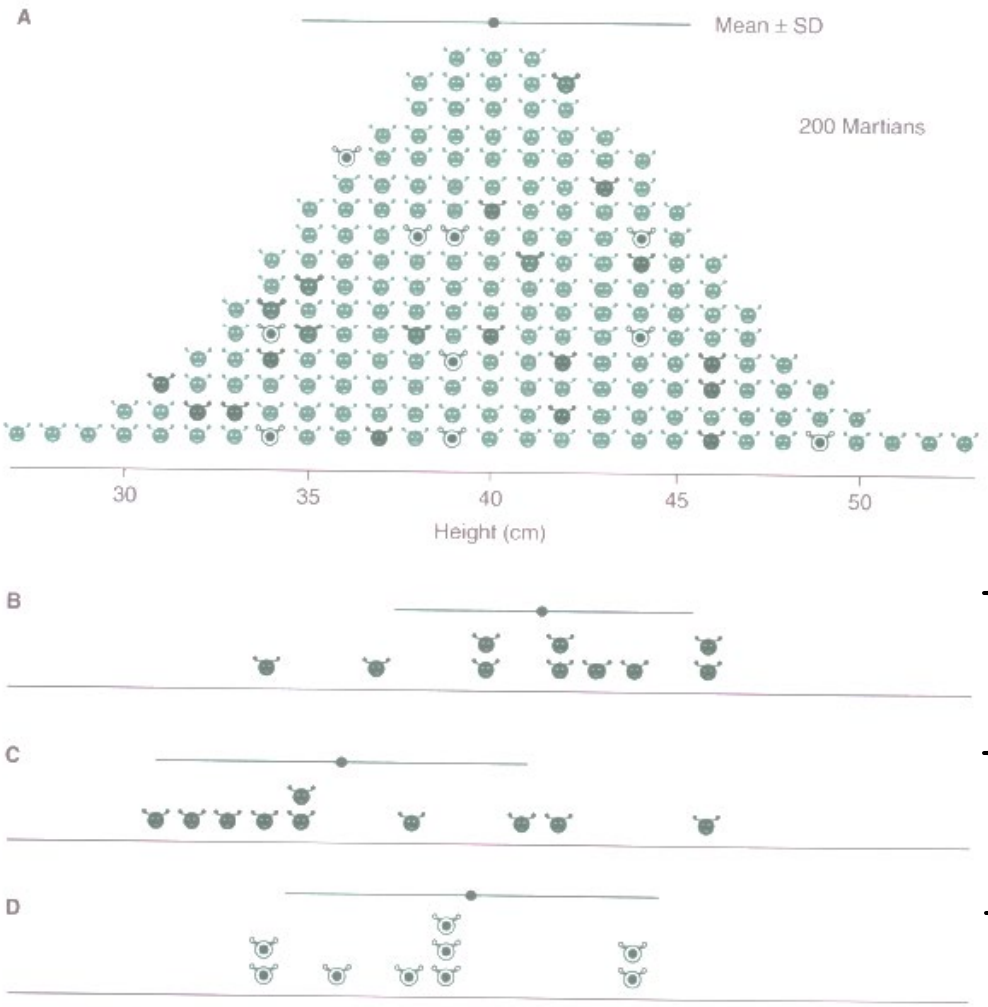


FIGURE 2-5. If one draws three different samples of 10 members each from a single population, one will obtain three different estimates of the mean and standard deviation.

$\mu = 40.$ cm

$\sigma = 5.0$ cm

$n = 10$

$\bar{x}_1 = 41.5$

$s_1 = 3.8$

$\bar{x}_2 = 36.$

$s_2 = 5.$

$\bar{x}_3 = 40.$

$s_3 = 5.$

• • •

Distribution of Sample Mean Values of Size $n = 10$

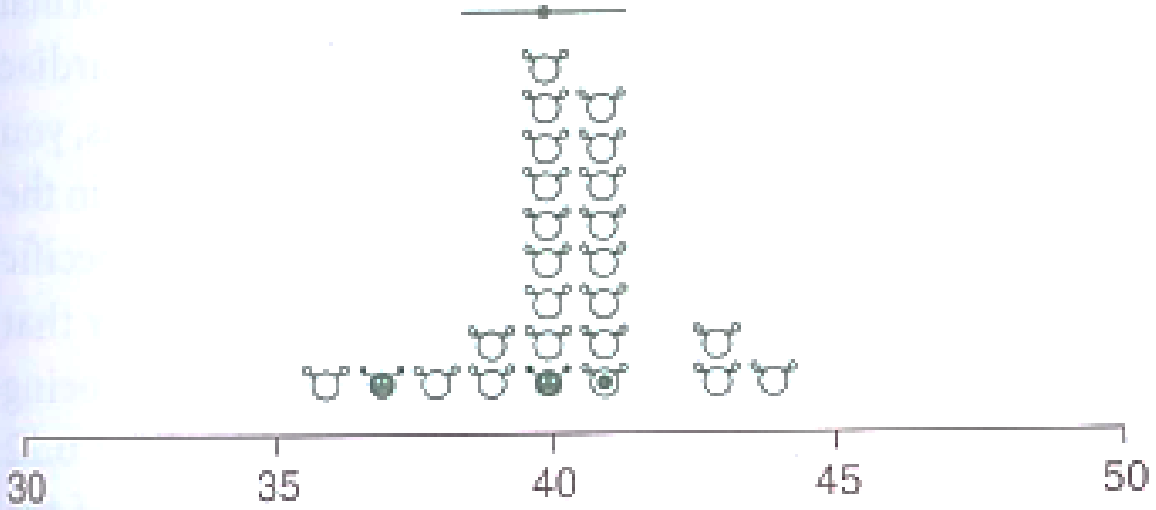
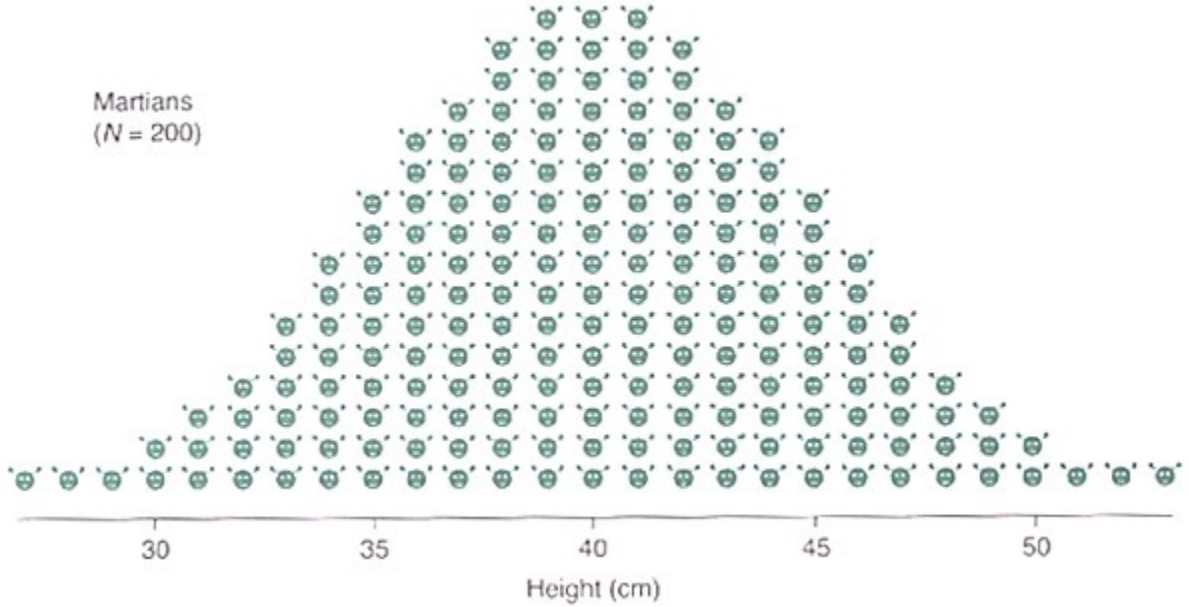


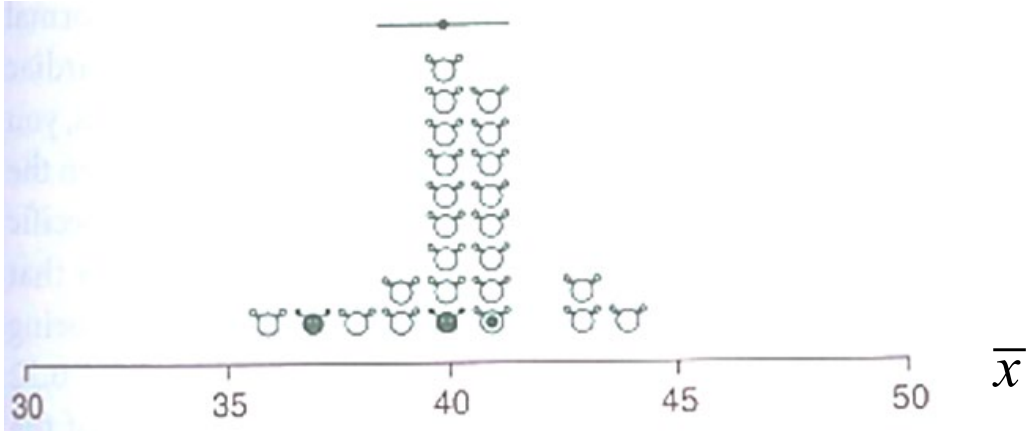
FIGURE 2-6. If one draws more and more samples—each with 10 members—from a single population, one eventually obtains the population of all possible sample means. This figure illustrates the means of 25 samples of 10 Martians each drawn from the population of 200 Martians shown in Figures 2-1 and 2-5A. The means of the three specific samples shown in Figure 2-5 are shown using corresponding symbols. This new population of all possible sample means will be normally distributed regardless of the nature of the original population; its mean will equal the mean of the original population; its standard deviation is called the standard error of the mean.



For Height

$\mu = 40.$ cm

$\sigma = 5.0$ cm



Mean Height (cm)

Sample mean of size 10

Sample mean of a sample of size n, called $\bar{x}$, is now a new random variable

Mean of sample means: $\mu_{\bar{x}}$

Standard deviation of sample means: $\sigma_{\bar{x}}$
(also called SEM: standard error of the mean)

$$\mu_{\bar{x}} = \mu$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

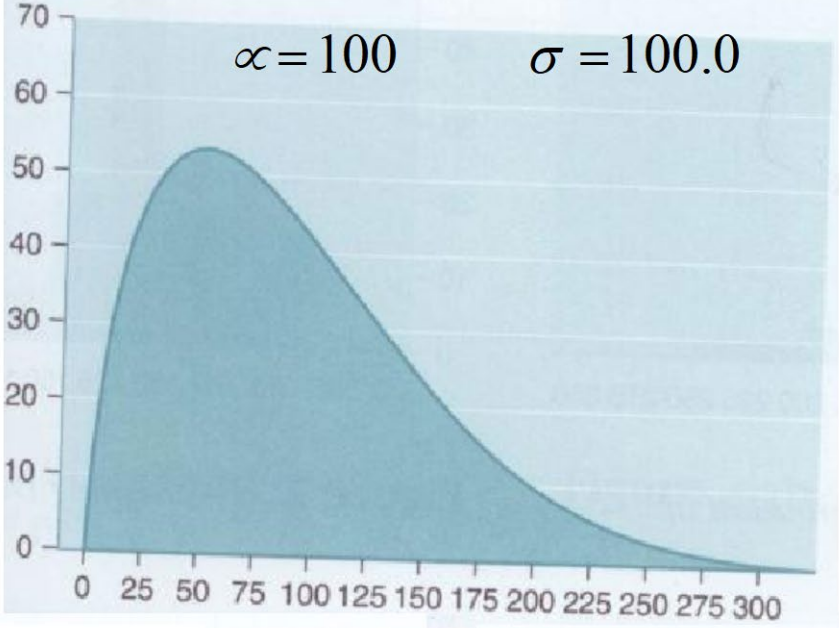
← Central Limit Theorem

Also, $\bar{x}$ is Normally distributed regardless of the distribution of x if n is "big" enough

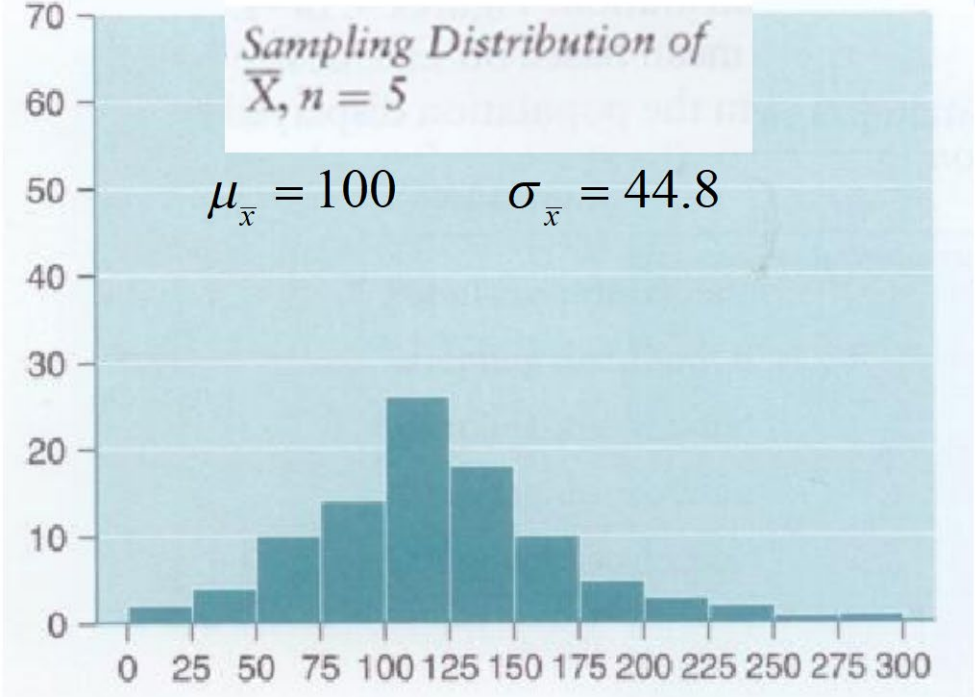
Idea of Central Limit Theorem:

- If a population has mean μ and variance σ^2
- If random samples of size $n \geq 30$ (with replacement) are taken from population
- Sampling distribution of sample means ($\bar{x}$) is approximately Normally distributed with $\mu_{\bar{x}} = \mu$ and $(\sigma_{\bar{x}})^2 = \sigma^2/n$, even if the underlying population is not Normally distributed

Figure 4.2 *Skewed Population*



Sampling Distribution of $\overline{X}$, $n = 5$



Sampling Distribution of $\overline{X}$, $n = 15$

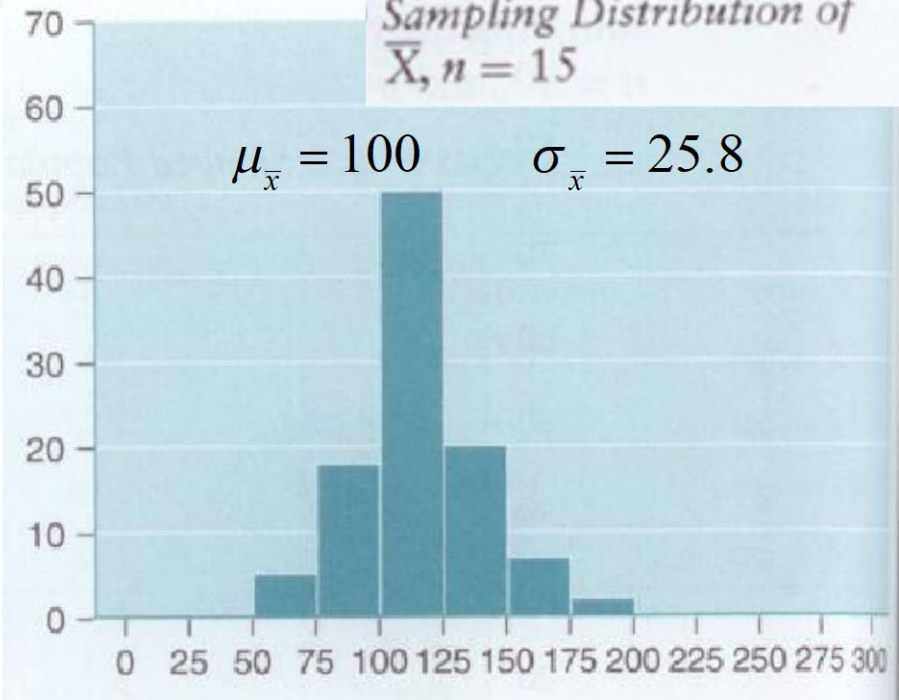


Figure 4.2c *Sampling Distribution of* $\overline{X}$, $n = 30$

$\mu_{\bar{x}} = 100$ $\sigma_{\bar{x}} = 18.2$

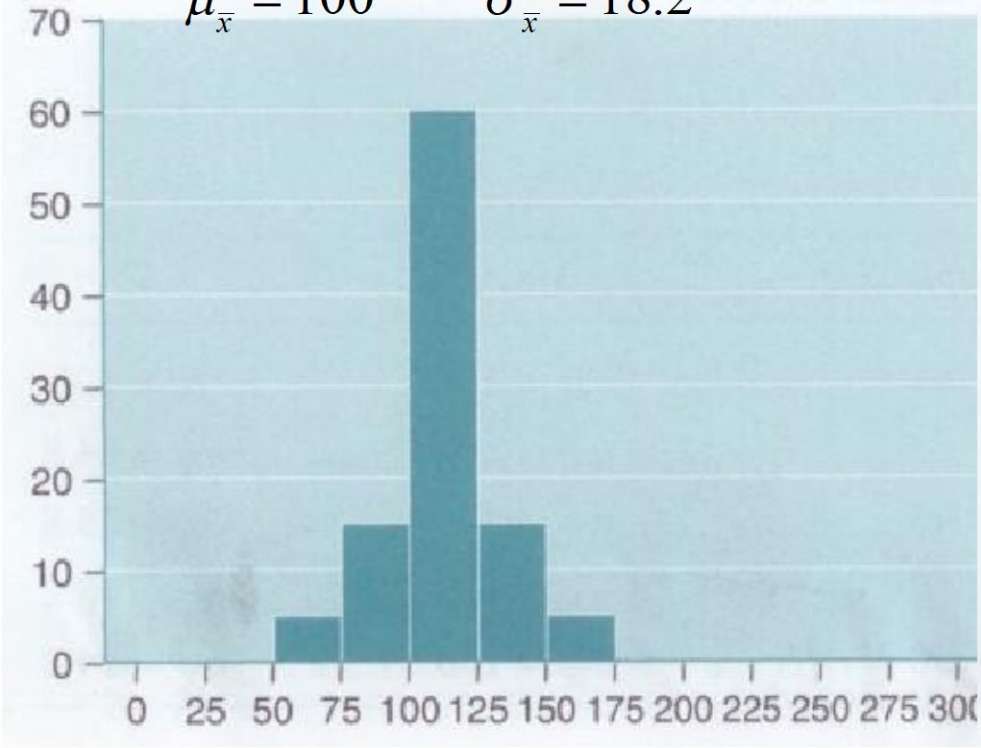
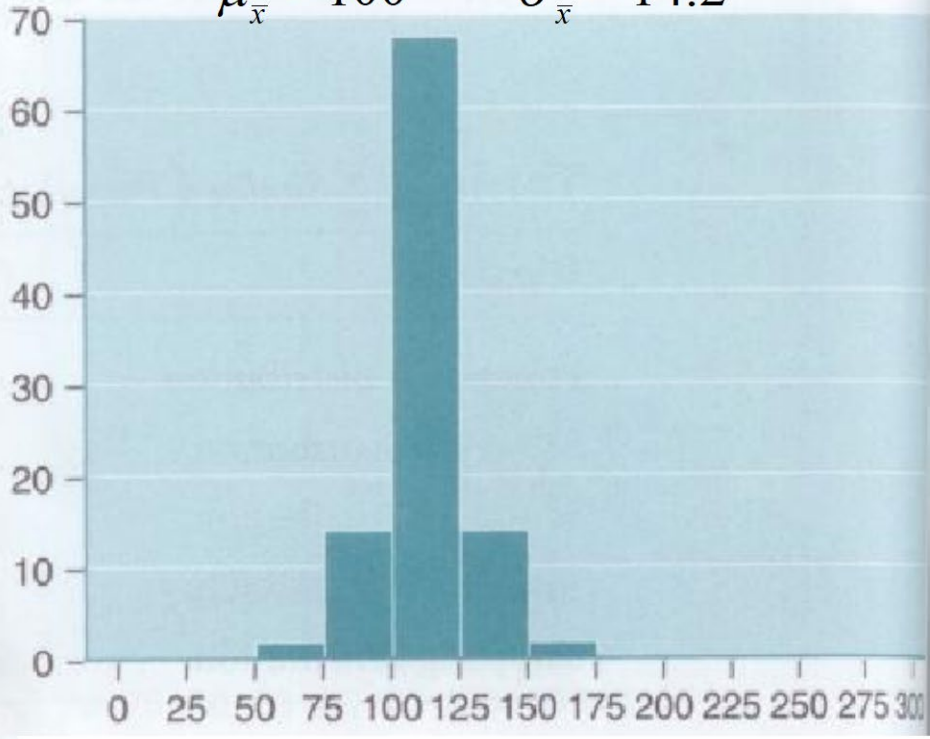


Figure 4.2d *Sampling Distribution of* $\overline{X}$, $n = 50$

$\mu_{\bar{x}} = 100$ $\sigma_{\bar{x}} = 14.2$



Properties of Sample Mean

$$E[\bar{x}] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} E[X] = \frac{1}{n}\sum_{i=1}^{n} \mu = \frac{1}{n}(n\mu) = \mu$$

$$Var(\bar{x}) = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n} \qquad \frac{\sigma}{\sqrt{n}} \leftarrow SEM$$

$$Var(x_{\text{median}}) > Var(\bar{x})$$

$$Var(x_{\text{avg:min,max}}) > Var(\bar{x})$$

Sample Mean $\bar{x}$ is an unbiased, minimum-variance estimator of μ

But how can we estimate $\sigma_{\bar{x}}$ from a single sample of size n?

Recall $\sigma_{\bar{x}}$

Best estimate of $\sigma_{\bar{x}}$ from a single sample of size n is

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

In Summary

Given a sample of size n: $x_i \ , \ i = 1, \ldots, \ n$

Sample mean: $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ Sample variance: $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2$

Standard error of mean: $s_{\bar{x}} = \frac{s}{\sqrt{n}}$ Standard error of variance: $s_{s^2} = s^2\sqrt{\frac{2}{n-1}}$

Summary of Some Statistical Methods to Test Hypotheses

Scale of measurement	Type of experiment: Two treatment groups consisting of different individuals	Three or more treatment groups consisting of different individuals	Before and after a single treatment in the same individuals	Multiple treatments in the same individuals	Association between two variables
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	z Test				
Nominal	Chi-square analysis-of-contingency table (Chapter 5)	Chi-square analysis-of-contingency table (Chapter 5)	McNemar's test (Chapter 9)	Cochrane Q†	Relative risk or odds ratio (Chapter 5)
					Logistic regression (slides)
Ordinal‡	Mann-Whitney rank-sum test (Chapter 10)	Kruskal-Wallis statistic (Chapter 10)	Wilcoxon signed-rank test (Chapter 10)	Friedman statistic (Chapter 10)	Spearman rank correlation (Chapter 8)
Survival time	Log-rank test or Gehan's test (Chapter 11)				

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†Not covered in this text.
‡Or interval data that are not necessarily normally distributed.

ANOVA: Summary

Experiment:

RV X is measured

m - # of treatment groups
n - # number of subjects per group

Null Hypothesis:

H_0: $\mu_1 = \mu_2 = \ldots = \mu_m$ Alternative hypoth: at least 2 $\mu's$ differ

1. Mean of each group

$\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_m$ $\left(\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i \right)$

2. Variance of each group

$s_1^2, s_2^2, \ldots, s_m^2$ $\left(s^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(x_i - \bar{x} \right)^2 \right)$

3. Calculate the within-groups est. of variance

$$s_{wit}^2 = \frac{1}{m}\sum_{i=1}^{m} s_i^2$$

4. Compute the overall (grand) mean

$$\bar{X} = \frac{1}{m}\sum_{i=1}^{m} \bar{x}_i$$

5. Compute standard error of the mean

$$s_{\bar{x}}^2 = \frac{1}{m-1}\sum_{i=1}^{m}\left(\bar{x}_i - \bar{X}\right)^2$$

6. Calculate the between-groups est. of variance

$$s_{bet}^2 = n s_{\bar{x}}^2$$

7. Compute the F Statistic

$$F = \frac{s_{bet}^2}{s_{wit}^2}$$

$\nu_n = m - 1$ Numerator degrees of freedom

$\nu_d = m(n-1)$ Denominator degrees of freedom

8. Determine F_{crit} (critical value of F)

$$F_{crit} \equiv F_{0.05}(\nu_n, \nu_d)$$ Obtained from F Table

In general $F_{crit} \equiv F_{\alpha}(\nu_n, \nu_d)$ α: significance level

9. Compare F to F_{crit} and State Conclusion

If $F > F_{crit}$ Reject null hypothesis with $P < \alpha$

Notes on ANOVA

- You are not responsible for the Sum of Squares formulation of the ANOVA test for the Midterm Exam
- Any ANOVA problem you might see on the Midterm Exam will have the same number of subjects in all groups (i.e., a balanced study design)

Summary of Some Statistical Methods to Test Hypotheses

Scale of measurement	Type of experiment: Two treatment groups consisting of different individuals	Three or more treatment groups consisting of different individuals	Before and after a single treatment in the same individuals	Multiple treatments in the same individuals	Association between two variables
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t Test Procedure (Student t Test)

Experiment:

RV X is measured — 2 treatment groups

n_1, n_2 # number of subjects in groups 1 & 2

Null Hypothesis:

$$H_0 : \mu_1 = \mu_2 \qquad H_1 : \mu_1 \neq \mu_2$$

1. Mean of each group

$$\bar{x}_1,\ \bar{x}_2 \qquad \left(\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i \right)$$

2. Variance of each group

$$s_1^2,\ s_2^2 \qquad \left(s_i^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(x_i - \bar{x} \right)^2 \right)$$

t Test Procedure (Student t Test)

3. Pooled Variance

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

4. Compute Value of t

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\nu = n_1 + n_2 - 2$$

Degrees of freedom

t Test Procedure (Student t Test)

5. Determine t_{crit} from t table

$$t_{crit} \equiv t_{\alpha}(\nu) \qquad \alpha\text{: significance level}$$

Using a two-tailed t table

6. Compare t to t_{crit} and State Conclusion

$$\text{If } |t| > t_{\alpha}(\nu) \quad \text{Reject null hypothesis with } P < \alpha$$

Assumptions of t Test

1. Data are interval data
2. Samples must be independent
3. Samples must be randomly selected
4. Population(s) must be Normally distributed
5. Variance(s) of each population must be equal

t Test for One-Sample Inference

Experiment:

RV X is measured n : number of subjects

Null Hypothesis:

$$H_0 : \mu = \mu_0 \ \left(\mu - \mu_0 = 0\right) \qquad H_1 : \mu \neq \mu_0$$

1. Mean of group

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

2. Variance of group

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \overline{x}\right)^2$$

3. Compute Value of t

Sample mean

Reference value against which you are comparing sample group

$$t = \frac{\bar{x} - \mu_0}{\sqrt{s^2 / n}}$$

$$\nu = n - 1$$

Degrees of freedom

4. Determine t_{crit} from two-tailed t table

$$t_{crit} \equiv t_{\alpha}(\nu)$$

α: significance level

5. Compare t and State Conclusion

If $|t| > t_{\alpha}(\nu)$ Reject null hypothesis with $P < \alpha$

Multiple Pairwise Comparisons using t Tests with Bonferroni or Holm Correction

Under H_0, best estimate of σ^2 is $s_{wit}{}^2$ from ANOVA (which was done first)

Degrees of freedom come with $s_{wit}{}^2$:

$$\nu = \nu_d = m(n-1) = N - m$$

if n is same for all groups

more general formula

$$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{\frac{s_{wit}^2}{n_A} + \frac{s_{wit}^2}{n_B}}}$$

Multiple Comparisons – Bonferroni *t* Test

Bonferroni inequality:

significance to use for each pairwise comparison

Overall level of Significance (e.g., 0.05)

$$\alpha_T < k\alpha$$

of pairwise comparisons

For each *t* Test comparison use:

$$\alpha = \frac{\alpha_T}{k}$$

Multiple Comparisons – Holm t Test

Idea: Use successively larger significance levels

significance to use for the jth pairwise comparison

$j = 1,\ldots,k$

Overall level of significance (e.g., 0.05)

$$\alpha_j = \frac{\alpha_T}{(k - j + 1)}$$

of pairwise comparisons

For example:

$k = 3$

$\alpha_1 = 0.0167$ largest value of $|t|$

$\alpha_2 = 0.025$:

$\alpha_3 = 0.05$ smallest value of $|t|$

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Estimating Proportions from Samples

$x_i,\ i = 1,\ldots,n \qquad x_i = 1$ in m subjects

sample proportion $\hat{p} = \dfrac{m}{n}$

sample variance $s^2 = \hat{p}(1-\hat{p})$

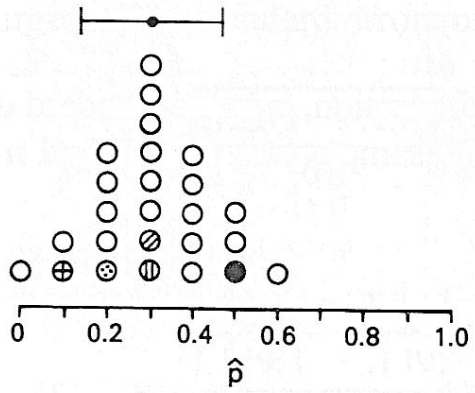


Figure 5-6 There will be a distribution of estimates of the proportion of left-footed Martians $\hat{p}_{left}$ depending on which random sample the investigator happens to draw. This figure shows the 5 specific random samples drawn in Figs. 5–4 and 5–5 together with 20 more random samples of 10 Martians each. The mean of the 25 estimates of p and the standard deviation of these estimates are also shown. The standard deviation of this distribution is the standard error of the estimate of the proportion $\sigma_{\hat{p}}$; it quantifies the precision with which $\hat{p}$ estimates p.

standard error of the proportion

$$s_{\hat{p}} = \sqrt{\frac{s^2}{n}}$$

$$s_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Summary of Some Statistical Methods to Test Hypotheses

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z Test

Table 5-1 Thrombus Formation in People Receiving Dialysis and Treated with Placebo or Aspirin

	Number of patients		
Sample group	Developed thrombi	Free of thrombi	Treated
Placebo	m_1 18	7	n_1 25
Aspirin	m_2 6	13	n_2 19
Total	24	20	44

Source: H. R. Harter, J. W. Burch, P. W. Majerus, N. Stanford, J. A. Delmez, C. B. Anderson, and C. A. Weerts "Prevention of Thrombosis in Patients on Hemodialysis by Low-Dose Aspirin," *N. Engl. J. Med.*, **301**:577–579, 1979. Reprinted by permission of the *New England Journal of Medicine.*

$$H_0 : p_{asp} = p_{pla}$$

Yates-Corrected *z* Test

(always use Yates's Continuity Correction for 2x2 contingency tables)

$$z = \frac{\left|\hat{p}_1 - \hat{p}_2\right| - \frac{1}{2}\left(1/n_1 + 1/n_2\right)}{\sqrt{\hat{p}\left(1-\hat{p}\right)\left(1/n_1 + 1/n_2\right)}} \qquad z \sim N(0, 1)$$

$$\hat{p}_1 = \frac{m_1}{n_1} \qquad \hat{p}_2 = \frac{m_2}{n_2} \qquad \hat{p} = \frac{m_1 + m_2}{n_1 + n_2}$$

$$z_{critical} = t_\alpha(\nu \to \infty) \qquad N(0, 1) = t(\nu \to \infty)$$

Using a two-tailed *t* table

$$\left|z\right| > z_{crit} = t_\alpha(\nu \to \infty) \text{ Reject } H_0 \qquad P < \alpha$$

Summary of Some Statistical Methods to Test Hypotheses

Scale of measurement	Type of experiment: Two treatment groups consisting of different individuals	Three or more treatment groups consisting of different individuals	Before and after a single treatment in the same individuals	Multiple treatments in the same individuals	Association between two variables
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Survival time	Log-rank test or Gehan's test (Chapter 11)				

χ² Test

Logistic regression (slides)

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More Than 2x2 Contingency Table

	Outcome				
Treatment	1	2	…	n_c	Total
Group 1	O_{11}	O_{12}	…		R_1
Group 2	O_{21}	O_{22}	…		R_2
…					…
Group n_r	O_{nr1}	…			R_{nr}
	C_1	C_2			n

$$E_{ij} = \frac{C_j}{n} R_i \qquad \chi^2 = \sum_{i=1}^{n_r} \sum_{j=1}^{n_c} \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}} \qquad v = \left(n_r - 1\right)\left(n_c - 1\right)$$

$$\chi^2 > \chi^2_\alpha(v) \qquad \text{Reject } H_0 \qquad P < \alpha$$

use Yates's correction only if 2x2:

$$\chi^2 = \sum\sum \frac{\left(\left|O - E\right| - \frac{1}{2}\right)^2}{E}$$

Use Bonferroni or Holm Correction to Control Overall Significance of Pairwise Comparisons

Order χ^2 values in descending order and compare to critical value

Calculated χ^2 value	j	α_j (Holm)	$\chi^2_{\alpha_j}(\nu=1)$
$\chi^2_{NvsN+P}=15.5$	1	$\frac{0.01}{(3-1+1)}=0.00\overline{333}$	$\chi^2_{0.00\bar{3}}<\chi^2_{0.001}=10.828$
$\chi^2_{PvsN+P}=12.4$	2	$\frac{0.01}{(3-2+1)}=0.005$	$\chi^2_{0.005}=7.879$
$\chi^2_{NvsP}=0.117$	3	$\frac{0.01}{(3-3+1)}=0.01$	$\chi^2_{0.01}=6.635$

k = number of pairwise comparisons = 3

$\alpha_T = 0.01 =$ *overall significance level*

for Holm correction, $\alpha_j = \frac{\alpha_T}{(k-j+1)}$

Comparing One Set of Categorical Observations Against Expected Frequencies: Goodness of Fit χ^2 Test

- Suppose we have a single set of categorical data and wish to compare the distribution of data points in each category to a set of expected frequencies (a "goodness of fit" test)
  The expected frequencies may come from a prior study/hypothesis
  There are N data points and k categories in this single data set

- H0: There is no difference between the observed and expected distribution of values in each category

OR

- H0: $p_{1,\text{Obs}} = p_{1,\text{Exp}}$, $p_{2,\text{Obs}} = p_{2,\text{Exp}}$, etc.
  (where p_i is the proportion of values in category i)

$$E_i = p_{i,\text{Exp}} \cdot N \qquad \chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \qquad \nu = k - 1$$

Comparing One Set of Categorical Observations Against Expected Frequencies: Goodness of Fit χ^2 Test

Example:

- A university surveyed graduating seniors about their experiences as undergrads
  - 60% reported getting no regular exercise during their undergraduate career
  - 25% reported getting sporadic exercise
  - 15% reported exercising regularly

- The university wished to increase the amount of exercise its students get while matriculating and launched a health-promotion campaign

- Four years later, the university surveyed the current graduating seniors and obtained the following results:

	No Regular Exercise	**Sporadic Exercise**	**Regular Exercise**	**Total**
Number of Students	255	125	90	470

Based on these data, is there a difference in students' exercise behavior after the campaign?

Example adapted from https://sphweb.bumc.bu.edu/otlt/mph-modules/bs/bs704_hypothesistesting-chisquare/index.html

Clinical Trials vs. Epidemiological Studies

A. Clinical Trials

Subjects are randomized to treatment groups and then the outcome is observed following assignment (prospective)

B. Epidemiological Studies

Subjects not randomized to treatment groups.

1) Prospective (cohort) – Enrolled in study then followed over time to determine outcome

2) Retrospective (case-control) – People who have the outcome of interest are identified and then grouped depending on whether they were exposed to the risk factor or not

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Relative Risk or Risk Ratio (RR)

The degree of association between nominal variables can be quantified by calculating the Relative Risk if the study is:

1) A Clinical Trail or
2) A Prospective Epidemiological Study

TABLE 5-13. Arrangement of Data to Compute Relative Risk

	Number of People		
Sample Group	Disease	No Disease	Total
Treated (or exposed to risk factor)	n_{TD}	n_{TN}	n_T
Control (or not exposed to risk factor)	n_{CD}	n_{CN}	n_C
Total	n_D	n_N	

Absolute risk reduction of event/disease (if proportion of control group greater than proportion of drug group):

$$\hat{p}_{placebo} - \hat{p}_{drug}$$

$$RR = \frac{Probability\ of\ event\ in\ treatment\ group}{Probability\ of\ event\ in\ control\ group} = \frac{n_{TD}/n_T}{n_{CD}/n_C} = \frac{\hat{p}_{drug}}{\hat{p}_{control}}$$

Number Needed to Treat (NNT)

- Number of patients that need to be treated in order for one patient to benefit
- Used when treatment increases likelihood of good outcome or decreases likelihood of bad outcome
- Calculated as the inverse of the absolute difference in prevalence of the outcome in the two groups:

$$NNT = \frac{1}{\left|\hat{p}_1 - \hat{p}_2\right|}$$

- Lower NNT means treatment is more effective; higher NNT means treatment is less effective

III. Number Needed to Harm (NNH)

- Number of patients that need to be treated/exposed in order for one patient to be harmed
- Used when exposure increases likelihood of bad/dangerous outcome or decreases likelihood of good outcome
- Calculated as the inverse of the absolute difference in prevalence of the (harmful) outcome in the two groups:

$$NNH = \frac{1}{\left|\hat{p}_1 - \hat{p}_2\right|}$$

- Lower NNH means treatment is more harmful; higher NNH means treatment is less harmful

- NNH – used as a measure of treatment safety

III. Number Needed to Harm (NNH)

- Number of patients that need to be treated/exposed in order for one patient to be harmed
- Used when exposure increases likelihood of bad/dangerous outcome or decreases likelihood of good outcome
- Calculated as the inverse of the absolute difference in prevalence of the (harmful) outcome in the two groups:

$$NNH = \frac{1}{\left|\hat{p}_1 - \hat{p}_2\right|}$$

- Lower NNH means treatment is more harmful; higher NNH means treatment is less harmful

- NNH – used as a measure of treatment safety

Odds Ratio (OR)

The degree of association between nominal variables can be quantified by calculating the Odds Ratio if the study is:

- A Retrospective Epidemiological Study
  (can also be used for either Prospective Study design)

TABLE 5-14. Arrangement of Data to Compute Odds Ratio

Sample Group	Number of People: Disease "Cases"	Number of People: No Disease "Controls"
Exposed to risk factor (or treatment)	n_{ED}	n_{EN}
Unexposed to risk factor (or treatment)	n_{UD}	n_{UN}
Total	n_D	n_N

Same calculation

$$OR = \frac{Odds\ of\ exposure\ in\ cases}{Odds\ of\ exposure\ in\ controls} = \frac{n_{ED}/n_{UD}}{n_{EN}/n_{UN}} = \frac{n_{ED}n_{UN}}{n_{UD}n_{EN}}$$

$$OR = \frac{Odds\ of\ cases\ in\ those\ exposed}{Odds\ of\ cases\ in\ those\ not\ exposed} = \frac{n_{ED}/n_{EN}}{n_{UD}/n_{UN}} = \frac{n_{ED}n_{UN}}{n_{UD}n_{EN}}$$

(prospective data only)

R Code and Outputs

- Know how to use `pnorm()` and `qnorm()` functions
- Know how to interpret outputs from
  - `aov()` function (for the ANOVA test; with formatting from `summary()` function, `print()` function, or similar)
  - `t.test()` and related functions (for various forms of the *t* Test)
  - `chisq.test()` and related functions (for various forms of the χ^2 Test)

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Midterm Exam: 90 minutes

Monday, 10:00 to 11:30 AM

You should bring the following items:

1) TWO 8.5" x 11" double-sided sheets of notes (anything you wish to put)
2) All statistical tables (WITHOUT notes)
   You can find them on the course's Brightspace page
   No tables will be distributed during the exam
   (z table – Normal Dist., F table, t table, Chi Sq. table)

3) Calculator (a phone is not an acceptable calculator)

All material through Session 13 (Descriptive Statistics through Measures of Association for Nominal Variables); not Session 14 lecture on Types of Errors/Power of Test

Review
- Reading from textbook (see syllabus)
- Lecture slides
- HW assignments
- Other questions at end of chapters

Exam: 6-8 Questions, 2-3 on probability